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Interfacial configurational entropy tuning strategy enabling liquid alloys for efficient depolymerization of polyolefin waste

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Accumulation of plastic waste presents a global environmental crisis, posing severe threats to ecosystems. Depolymerization offers a promising route for closed-loop recycling of plastic waste. Nevertheless, depolymerizing chemically stable polyolefins remains challenging, yet fails to achieve high light olefin yields and stability. Herein, we propose an interfacial configurational entropy tuning strategy for liquid alloy catalysts, inducing active site interfacial enrichment and optimizing electronic structure. With this strategy, a quaternary GaInNiSn catalyst with medium entropy is developed for recycling polyolefins into light olefins under atmospheric pressure without co-reactant consumption, achieving a space-time yield of 181.5 mmol g_{cat}^{-1} h^{-1}. Mechanistic insights reveal that $Ni^{\delta-}$–$Sn^{\delta+}$ active sites form within the dynamic Ga interface, enabling C–H bond activation to promote β-scission. Furthermore, a photovoltaics-driven depolymerization system is constructed, maintaining light olefin output of 52.8 L h^{-1} over 120 hours. Our work unlocks a potent tool for efficient and sustainable plastic waste utilization.

Plastics, the most extensively used synthetic materials in modern society, have revolutionized human civilization through their exceptional durability, versatility, and modifiability[1,2]. It is anticipated that the annual global plastic production will reach 1.2 billion tons by 2050, with a staggering accumulation of waste plastics of 25 billion tons[3,4]. However, plastics exhibit resistance to natural degradation and currently lack cost-effective recycling solutions. Globally, less than 10% of waste plastic undergoes proper recycling, and the overwhelming majority of discarded plastic contributes to severe environmental degradation and substantial resource depletion[5–7]. This striking disparity between the explosive growth in plastic production and the persistently low recycling rate threatens to trigger incalculable ecological crises[8]. Accordingly, the implementation of systematic waste plastic recycling is critically urgent, particularly for polyolefins, which constitute over 77% of global plastic production and dominate the waste stream[3,9]. At present, physical recycling is largely confined to single-category, clean, and high-value waste plastics; in stark contrast, chemical recycling circumvents these limitations by deconstructing polymers at the molecular level to regenerate high-quality products[10,11]. Although chemical recycling technology is well-established for easily recoverable polyesters like polyethylene terephthalate (PET), its application to chemically stable polyolefins, such as polyethylene (PE) and polypropylene (PP), remains a formidable challenge[12,13].

Depolymerizing polyolefins into monomers, chemically identical to petroleum-derived precursors, represents the most promising approach among chemical recycling technologies[14,15]. These reclaimed monomers can be directly repolymerized into new plastic products or utilized as platform chemicals in value-added industrial processes. Nonetheless, depolymerization of polyolefin waste is hindered by the

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inherent inertness of C–C and C–H bonds, necessitating harsh conditions to overcome kinetic and thermodynamic barriers[16–18]. Moreover, efficient depolymerization entails not only cleaving these inert bonds but also precise control over the cleavage process to avoid random dissociation[19,20]. Existing depolymerization approaches, consequently, are typically energy-intensive and involve significant efforts to improve product yield[21,22]. In this context, pulsed electrified heating offers a highly competitive process for polyolefin depolymerization[23,24]. This process utilizes transient heating to overcome kinetic barriers, followed by cooling, shifting the depolymerization reaction away from chemical equilibrium and suppressing side reactions. Despite these merits, the pronounced asymmetry between heating and cooling durations, in which cooling requires nine times longer than heating, drastically reduces the space-time yield (6.57 mmol C_3H_6 cm^{-2} h^{-1}). Disruptively, the tandem catalytic cracking and isomerizing ethenolysis process (CIE) ingeniously integrates olefin isomerization and olefin metathesis, enhancing monomer selectivity to attain a high C_3H_6 space-time yield of 135.5 mmol g_{cat}^{-1} h^{-1} in polyolefin depolymerization, representing the highest level attained to date[25,26]. Nevertheless, this process requires elevated pressures and near-stoichiometric consumption of the C_2H_4 co-reactant. The operational stability of these depolymerization processes, additionally, requires further enhancement, with current continuous operation durations less than five hours[23,26]. Therefore, it is imperative to develop depolymerization processes integrating high space-time yield, outstanding monomer selectivity, robust operational stability, and cost-effectiveness.

Herein, we report a catalytic depolymerization process using induction-heated liquid alloys (Fig. 1), converting polyolefin waste into light olefins at atmospheric pressure without requiring co-reactants. Central to this process is a quaternary GaInNiSn liquid alloy catalyst (LAC, Supplementary Table 1), which is designed by an interfacial configurational entropy ($S_{\text{if-config}}$) tuning strategy (Supplementary Note 1). We incorporate three elements into the dominant Ga liquid metal to induce atomic disorder at the alloy interface, thus forming a medium-entropy liquid alloy GaInNiSn (MELAC) with a $S_{\text{if-config}}$ of 1.16 R. In stark contrast to the binary or ternary liquid alloy catalysts, the increase in the $S_{\text{if-config}}$ of MELAC attenuates interfacial Gibbs free energy (Eq. S8), thereby modulating Ni active sites and promoting their enrichment on the Ga interface. Leveraging induction heating for rapid depolymerization, the process significantly suppresses low-temperature side reactions and facilitates PP depolymerization, delivering a space-time yield (STY) of 181.5 mmol g_{cat}^{-1} h^{-1} with a selectivity of 79.7%. Mechanistic insights suggest that Sn electronically modifies Ni to form $Ni^{\delta-}$–$Sn^{\delta+}$ sites at the GaIn interface, activating C–H bonds for selective depolymerization via β-scission. Furthermore, we

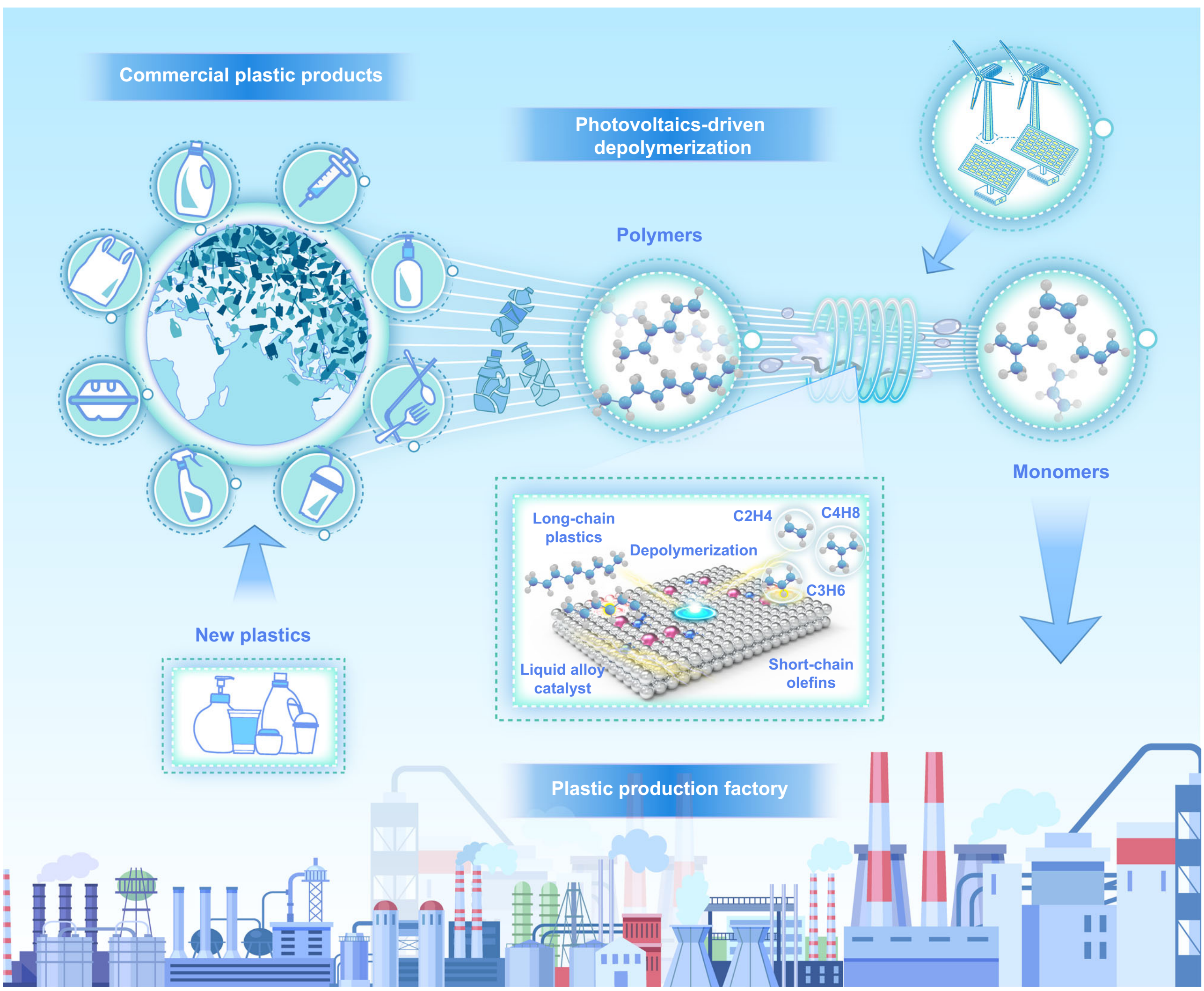

Fig. 1 | Closed-loop recycling strategy for waste plastics. Schematic illustration of typical commercial plastic products entering the waste stream, the proposed outdoor photovoltaics-driven depolymerization system for plastic waste, and the potential mechanism for depolymerization of plastic waste into light olefins using the GaInNiSn MELAC powered by induction heating.

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constructed an outdoor photovoltaics-driven depolymerization system for real-world polyolefin waste, maintaining a light olefin production rate of 52.8 L h^{-1} in more than 120 h of continuous operation. This system holds promise for the large-scale depolymerization of waste polyolefin plastics into light olefins, contributing to sustainable resource utilization and net-zero carbon emissions.

Results

Synthesis and characterization of GaInNiSn MELAC

The quaternary GaInNiSn LAC was synthesized by the 'one-pot' method (Supplementary Fig. 1). High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) analysis showed that the GaInNiSn MELAC maintained a homogeneous liquid phase state at room temperature, with no discernible presence of metal or metal oxide nanoparticles (Supplementary Fig. 2a, b). Energy dispersive X-ray spectroscopy (EDS) mapping (Fig. 2a) disclosed a uniform distribution of the active metal Ni and the regulatory metal Sn within the Ga solvent matrix. Inductively coupled plasma-optical emission spectrometry confirmed a Ni-to-Sn molar ratio of 3:1, corresponding to weight percentages of approximately 6.01 wt% Ni and 4.07 wt% Sn (Supplementary Table 2). The Hall effect measurement revealed that the GaInNi LELAC and GaInNiSn MELAC exhibit excellent electrical conductivity (Supplementary Table 3), serving as the receptor of induction heating. Contact angle tests (Supplementary Fig. 3) exhibited that Sn incorporation remarkably reduced the surface tension of GaIn alloys, thus providing a large solid-liquid contact area for plastic depolymerization. In-situ X-ray photoelectron spectroscopy (XPS) analysis corroborated the presence of Ni, Sn, Ga, and In in the GaInNiSn (Supplementary Figs. 4–6), while the GaInNi LAC did not exhibit distinct peaks of Ni 2*p* signals (Fig. 2b), suggesting that the incorporation of Sn atoms promotes the enrichment of Ni atoms at the alloy surface. Notably, the GaInNiSn LAC exhibited a $S_{\text{if-config}}$ of 9.62 J·mol^{-1} K^{-1} (1.16 *R*) and resided within a medium-entropy range spanning 1.0 *R* to 1.5 *R*. In stark contrast to the binary or ternary LAC, the $S_{\text{if-config}}$ of quaternary GaInNiSn was remarkably greater than those of low-entropy liquid alloys (LELAC) (Fig. 2c). This attribute endows the GaInNiSn MELAC with low interfacial Gibbs free energy (Supplementary Note 1), thus facilitating interfacial enrichment and reconstruction of active sites[27]. Ab initio molecular dynamics calculations (Supplementary Fig. 7) further disclosed that the Sn atom dragged the Ni atom outward to the surface of GaIn LELAC on account of the strong Sn–Ni interaction, attenuating the surface tension of the GaInNi LELAC. This phenomenon manifests that the introduction of Sn increases the $S_{\text{if-config}}$ at the interface by inducing atomic disorder, thus disrupting the original coordination environment and forming Sn–Ni interactions[28,29]. Moreover, the robust Sn–Ni interaction reduces the energy barrier for Ni migration to the surface and overcomes the solvation constraints imposed by the surrounding Ga atoms[30,31]. Since the depolymerization process by LAC is a typical solid-liquid heterogeneous catalysis reaction, the Ni surface enrichment in the GaInNiSn MELAC increases the specific active surface area, leading to superior performance in bond cleavage. Furthermore, the GaInNiSn MELAC induces a distinct shift of the Ni 2*p* peak to low binding energy, accompanied by a concurrent shift of the Sn 3*d* peak to high binding energy (Supplementary Figs. 4, 5), indicating potential electron donation characterized by $Ni^{\delta-}$–$Sn^{\delta+}$ polarization. Notably, despite the elevated temperature of 500 °C augments Ga solvent mobility, the increase in the $S_{\text{if-config}}$ inhibits the agglomeration of active Ni sites and their diffusion into the Ga bulk phase (Supplementary Table 1). Meanwhile, the strong Sn–Ni electronic interaction dynamically anchors the $Ni^{\delta-}$ atoms at the interface, preventing their encapsulation by surrounding Ga or In atoms.

X-ray absorption spectroscopy (XAS), furthermore, uncovered the electron state and precise local coordination of Ni atoms in the GaInNiSn MELAC. The X-ray absorption near-edge structure (XANES) spectrum of Ni *K*-edge (Fig. 2d) exhibited a pronounced negative absorption edge shift for the GaInNiSn MELAC relative to Ni foil, contrasting with the negligible change in the GaInNi LELAC. This shift confirms substantial electron accumulation at Ni atoms in the Sn-incorporated system, whereas the Sn-free system retains Ni atoms in a near-zero valence state. The Sn *K*-edge XANES spectrum (Fig. 2e) showed an absorption edge in the GaInNiSn MELAC positioned between Sn foil and SnO_2, suggesting a partial positive charge ($Sn^{\delta+}$, $0 < \delta < 4$) with reduced electron density. The electronic states of $Ni^{\delta-}$ and $Sn^{\delta+}$ corroborate the charge transfer from Sn to Ni, forming the $Ni^{\delta-}$–$Sn^{\delta+}$ sites in the GaInNiSn MELAC, as identified by in-situ XPS and charge density difference (Supplementary Fig. 8). To obtain an accurate coordination configuration for the GaInNiSn MELAC, Fourier transform (FT) k^2-weighted extended X-ray absorption fine structure (EXAFS) spectra were collected. The Ni *K*-edge FT-EXAFS spectra showed that the first peak in the GaInNiSn MELAC corresponded to the Ni–Ga path, closely matching the Ni–Ga peak in the GaInNi LELAC (Supplementary Fig. 9). Furthermore, a Ni–Ga peak for the GaInNiSn MELAC at $(R + \alpha) = 2.08$ and $k = 6.64$ in Wavelet transform (WT) analysis resembled the Ni–Ga bond in the GaInNi LELAC but strikingly deviated from the Ni–Ni bond in Ni foil (Fig. 2h and Supplementary Figs. 10a, 11), indicating an atomic dispersion of Ni species in the GaInNiSn MELAC. Likewise, both FT-EXAFS (Fig. 2f) and WT spectra (Fig. 2i and Supplementary Fig. 10b) of the Sn *K*-edge illustrated the Sn–Ni/Ga scattering paths, which are difficult to distinguish given their similarity in *k* and *R* space. Furthermore, various scattering models were employed to fit the EXAFS curves (Supplementary Table 4) to identify the coordination environment of the GaInNiSn MELAC. In terms of the selection criteria[32], Model 5 emerged as the most possible coordination configuration featuring the Sn–Ni scattering path in the first coordination shell, lending evidence to the presence of Ni–Sn interactions (Fig. 2g).

Catalytic depolymerization of polyolefin waste

Generally, reaction temperature and duration strongly influence the depolymerization process. At temperatures below 500 °C, incomplete C–C bond cleavage reduced the selectivity and STY of light olefins, whereas increasing the temperature to 600 °C induced deep cracking and aromatization, generating substantial CH_4 and aromatic byproducts that further deteriorate the STY of light olefins (Supplementary Fig. 13)[16,21]. Moreover, because the polypropylene was completely depolymerized within 3 min, further extension of the reaction time led to a decrease in the STY of light olefins to 129.9 mmol g_{cat}^{-1} h^{-1} (Supplementary Fig. 14). In stark contrast to a conventional resistance furnace, induction heating provides rapid (heat-up rate > 100 °C s^{-1}), uniform (surface temperature variation of ±5 °C), and precise (targeting the LAC directly) reaction conditions for polyolefin depolymerization (Supplementary Figs. 13–18)[23]. Notably, the processing time for polypropylene plastics under induction heating was approximately 3.0 minutes, while the equivalent quantity of plastic required over 35.5 minutes for processing in a resistance furnace (Supplementary Fig. 19 and Supplementary Table 5). By achieving a swift rise to the target depolymerization temperature, induction heating minimizes the exposure to low temperatures, thereby suppressing partial chain scission and random cleavage. It is observed that, under induction heating, residue weight percentage and CH_4 selectivity were remarkably lower than those heated by a conventional resistance furnace (Supplementary Tables 5, 6). Compared with this technique, Joule heating generally induces localized thermal inhomogeneities, triggering excessive side reactions (*e.g.*, coking) that reduce the selectivity of the target product[24]. Microwave heating is constrained by the low dielectric loss of polyolefins, resulting in suboptimal energy efficiency and difficulty in controlling heating uniformity and reaction processes[33–35]. Furthermore, the incorporation of Sn as a promoter into the GaIn LELAC improved the C_3H_6 selectivity and STY in polypropylene depolymerization (Supplementary Fig. 12 and Supplementary Table 6). This is because Sn can reduce the surface tension of the

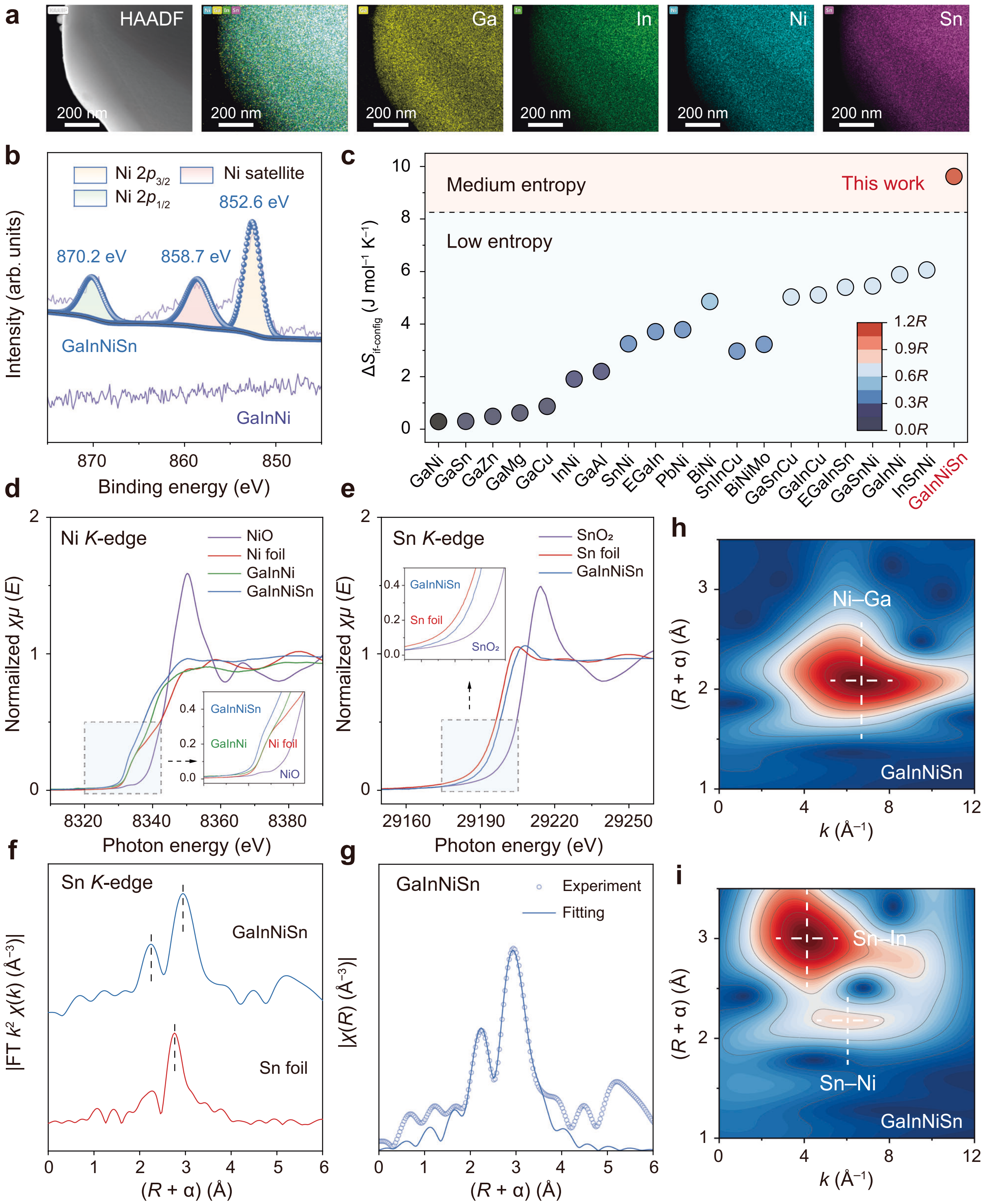
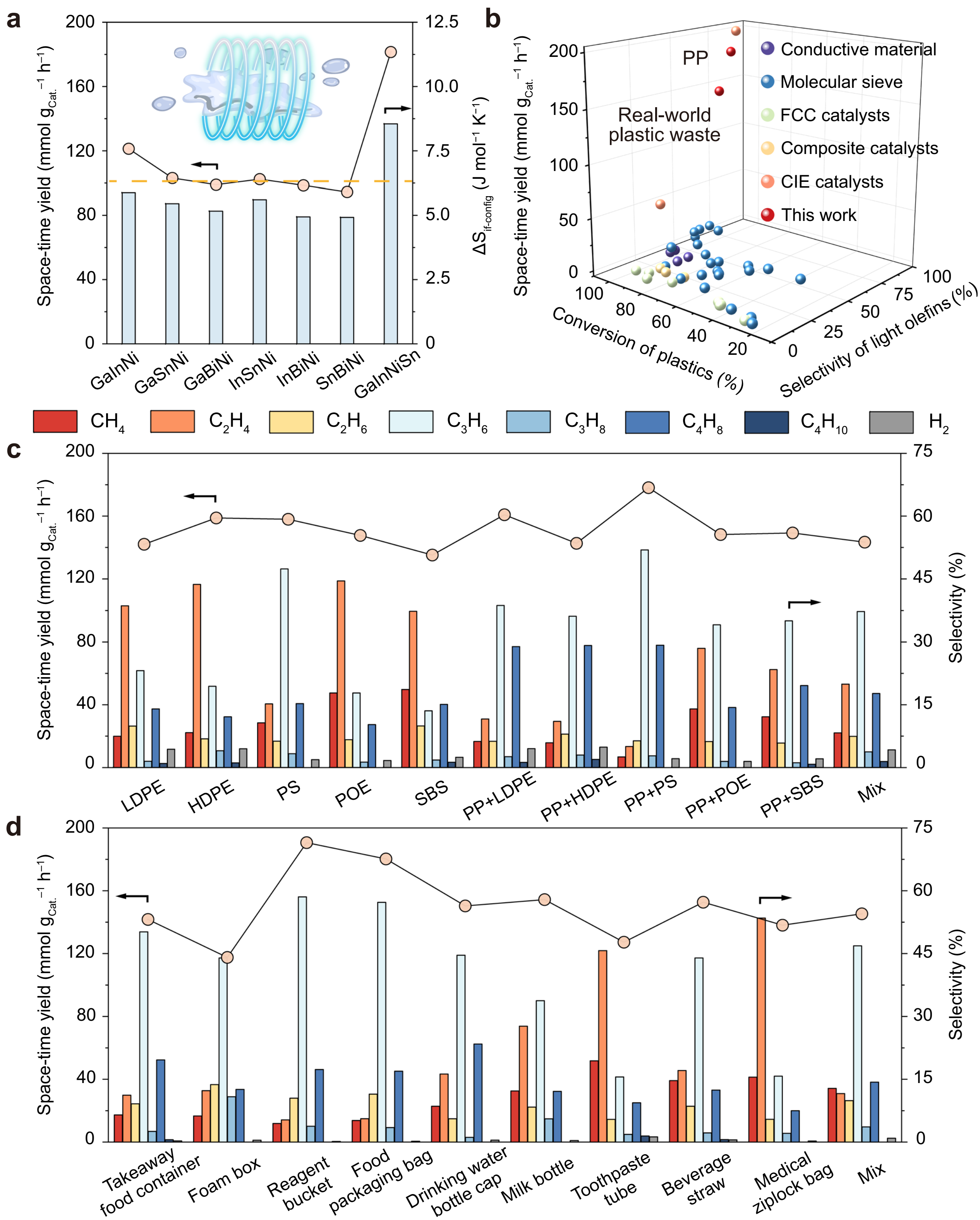


Fig. 2 | Characterizations of the GaInNiSn MELAC. **a** HAADF-STEM image and corresponding EDS mapping of the GaInNiSn MELAC. **b** In-situ XPS spectra of the GaInNi LELAC and the GaInNiSn MELAC. **c** Comparison of the $S_{\text{if-config}}$ of liquid alloys between this work and previous literature[30,31,49–55]. **d** Ni *K*-edge XANES spectra of the GaInNiSn MELAC and reference samples. The inset shows the local enlargement of Ni *K*-edge XANES spectra. **e** Sn *K*-edge XANES spectra of the GaInNiSn MELAC and reference samples. The inset shows the local enlargement of Sn *K*-edge XANES spectra. **f** FT k^2-weighted EXAFS spectra of the GaInNiSn MELAC and reference samples. **g** First-shell and second-shell fitting of Sn *K*-edge FT-EXAFS spectra for the GaInNiSn MELAC. **h**, **i** WT for the k^2-weighted EXAFS signals of the GaInNiSn MELAC.



LAC, as evidenced by the characterization section, and thereby expand the contact area between the liquid alloy and the solid plastic, facilitating an increase in available depolymerization sites of LACs[36]. Meanwhile, incorporating Ni, with strong C–H bond-activation capability, into GaIn LELAC is expected to improve depolymerization performance. Nonetheless, the cage effect impedes effective distribution of Ni to the solid-liquid reaction interface, leading to a modest increase in the depolymerization process. Once Sn and Ni are incorporated into the GaIn LELAC, the polyolefin depolymerization process is significantly enhanced. At a reaction temperature of 500 °C, the GaInNiSn MELAC achieved a light olefin selectivity of 79.7% and a STY of 181.5 mmol g_{cat}^{-1} h^{-1}, reaching 150% and 200% increases compared with those of the GaIn LELAC, respectively (Fig. 3a). Notably, 96.0% of the plastic were depolymerized (Supplementary Table 6),

Fig. 3 | Depolymerization performance of the GaInNiSn MELAC. a Comparison of depolymerization performance between ternary and quaternary liquid alloy catalysts (all active metals being Ni). The dashed line denotes the average STY of the ternary LACs (103.5 mmol g_{cat}^{-1} h^{-1}). Reaction conditions: 1.00 g of polypropylene plastic, 1.95 g of different LACs (SnBiNi, InBiNi, InSnNi, GaBiNi, GaSnNi, GaInNi, and GaInNiSn), N_2 feed gas at a flow rate of 4 mL min^{-1}, and 500 °C for 3 min. **b** Comparison of the performance of different depolymerization and recycling technologies, including the conversion of plastic, the selectivity of light olefins, and the space-time yield of light olefins[23–26,44,56–64]. **c** Space-time yield of light olefins and selectivity of gaseous products during depolymerization of diverse polyolefin plastics. Reaction conditions: 1.00 g of reactants (PP, LDPE, HDPE, PS, POE, and SBS), 1.95 g of GaInNiSn MELAC, N_2 feed gas at a flow rate of 4 mL min^{-1}, and 500 °C for 3 min. PP + LDPE, PP + HDPE, PP + PS, PP + POE, and PP + SBS, respectively, denote equal mass ratio binary blends of PP and the corresponding polymer. Mix denotes the equal mass ratio mixture of these six reactants. **d** Space-time yield of light olefins and selectivity of gaseous products during depolymerization of real-world plastic waste. Reaction conditions: 1.00 g of reactants (food packaging bags, take-away containers, drinking water bottle caps, milk bottles, beverage straws, toothpaste tubes, foam boxes, reagent buckets, and medical ziplock bags), 1.95 g of GaInNiSn MELAC, N_2 feed gas at a flow rate of 4 mL min^{-1}, and 500 °C for 3 min. Mix denotes the equal mass ratio mixture of these nine reactants.

while the residual product was mainly composed of jet fuel hydrocarbons (C_8–C_{16}) with minimal paraffins (C_{22+}) (Supplementary Figs. 20, 21). As shown in Fig. 3a, although the $S_{\text{if-config}}$ of the quaternary LAC shows a modest increase in comparison with ternary LACs, the quaternary GaInNiSn LAC yields an average improvement of 75.4% in the depolymerization performance in terms of STY of light olefins, highlighting the effectiveness of this $S_{\text{if-config}}$ tuning strategy. When benchmarked against state-of-the-art catalysts, the GaInNiSn LAC is far ahead in terms of the STY and monomer selectivity (Fig. 3b and Supplementary Table 7). Notably, the STY of light olefins for the GaInNiSn MELAC (1 bar) is threefold that of CIE under atmospheric pressure (1 bar), even approaching the STY under high pressures (15 bar).

The applicability of the catalytic depolymerization process using the GaInNiSn MELAC under induction heating was validated using diverse pure polyolefins (Fig. 3c and Supplementary Table 8). Low-density PE (LDPE) and high-density PE (HDPE) were depolymerized to their monomer of C_2H_4, with selectivity of 38.6% and 43.7%, respectively. Meanwhile, polystyrene (PS) depolymerization yielded valuable short-chain olefins with a total selectivity of 77.9%, demonstrating that our strategy can even cleave the stable aromatic ring structure of PS. The GaInNiSn MELAC was also applied to structure-complex thermoplastic elastomers: block copolymer of ethylene-octene (POE) and triblock copolymer of styrene-butadiene-styrene (SBS). Strikingly, both POE and SBS were predominantly deconstructed to C_2H_4, with total olefin selectivity of 72.6% and 65.9%, respectively, underscoring the versatility of our strategy. Furthermore, the depolymerization of mixed pure polyolefin streams, which are challenging to separate due to their comparable densities, was investigated[37,38]. The depolymerization of pure polyolefin blends, specifically PP + LDPE, PP + HDPE, and PP + PS (1:1 mass ratio), yielded light olefin selectivities of 76.3%–86.2% and STYs of 142.7–178.2 mmol g_{cat}^{-1} h^{-1}. Besides, the product distribution is susceptible to the composition of the blends. For instance, blends containing PP with non-block copolymers (PE or PS) primarily yielded C_3H_6 and C_4H_8, whereas those containing PP with block copolymers (POE or SBS) predominantly generated C_2H_4 and C_3H_6. The depolymerization of a six-component blend, comprising equal masses of the above pure polyolefins, produced light olefins with a selectivity of 74.9% and a STY of 143.4 mmol g_{cat}^{-1} h^{-1}, still remarkably outperforming other technologies (Fig. 3b). The depolymerization capability of the GaInNiSn MELAC was further evaluated using real-world plastic wastes. These complex waste streams comprise a variety of polymers, additives, and contaminants (e.g., food residues, inks, and labels), which significantly exacerbate recycling challenges[39,40]. Representative post-consumer plastic products from daily life were selected for testing, including food packaging bags, takeaway containers, milk bottles, reagent buckets, foam boxes, toothpaste tubes, and medical ziplock bags (Supplementary Fig. 22). As illustrated in Fig. 3d and Supplementary Table 9, the GaInNiSn MELAC depolymerizes these untreated, commercially sourced plastic items into light olefins, with the selectivities approaching that obtained from pure polymers, ranging from 70.7% to 84.3%, despite their complex compositions and unknown contaminants. Notably, even when the aforementioned post-consumer plastic wastes were physically mixed in equal mass ratios to simulate a heterogeneous real-world waste, the depolymerization products achieved a light olefin selectivity of 72.8% with a STY of 145.3 mmol g_{cat}^{-1} h^{-1}, comparable to those of the pure polyolefin blends (Fig. 3c). These results demonstrate the robustness of our strategy in directly depolymerizing real-world, contaminated, and blended polyolefin wastes into valuable light olefins without prior sorting and cleaning.

Thermogravimetric analysis-Fourier transform infrared spectroscopy-mass spectrometry (TGA-FTIR-MS) was conducted to reveal the performance enhancement mechanism. Notably, the reaction rate of the GaInNiSn MELAC was greater than that of the other samples on account of the increased solid-liquid contact area between plastics and LAC (Supplementary Fig. 23). The formation of =C–H peaks at low temperatures manifests that the interface-enriched $Ni^{\delta-}$–$Sn^{\delta+}$ sites contribute to C–H bond activation (Fig. 4a, b and Supplementary Figs. 24, 25). The negatively charged Ni site in the GaInNiSn MELAC leads to strong H adsorption (Supplementary Fig. 26), reducing C–H activation barriers and boosting subsequent β-scission[41]. Furthermore, at 360–400 °C, the peak areas of =C–H and C=C bonds in the GaInNiSn MELAC were 2.6 and 2.3 times greater than those in the GaInNi LELAC, respectively (Fig. 4c), attesting to the robust capability of the $Ni^{\delta-}$–$Sn^{\delta+}$ sites to facilitate the β-scission of carbocation. Unlike other samples, light olefins and alkanes were simultaneously generated during the depolymerization process in the GaInNiSn MELAC (Fig. 4d, e, and Supplementary Fig. 27), further confirming the prevalence of β-scission. Generally speaking, the generation of short-chain hydrocarbons from PP pyrolysis primarily derives from the random cleavage of C–C bonds (Supplementary Fig. 28)[38]. In contrast, during the catalytic depolymerization of PP over the GaInNiSn MELAC, C–H bonds are first activated to form carbocation intermediates that subsequently undergo β-scission to produce light olefins (Fig. 4f)[42,43].

Photovoltaics-driven depolymerization system

To assess the scalability of this process for practical implementation, we constructed an outdoor photovoltaics-driven depolymerization system in Dalian City, Liaoning Province, China. As illustrated in Fig. 5a, the system comprises four photovoltaic panels (irradiation area of 10.3 m^2, module efficiency of 24%), an electromagnetic induction heating apparatus (operating power 380 Wh), a vacuum heat pipe reactor, and four storage batteries with a combined capacity of 9.6 kWh. This system was loaded with 11.7 g of the GaInNiSn MELAC to recycle real-world polyolefin waste, consistent with the blends tested in Section 2.2. The real-world waste fragments are conveyed into the reactor via a continuous feed apparatus at a feed rate of 2.92 g min^{-1} (Supplementary Fig. 29). The process flow and unit schematics of the system are provided in Fig. 5b, with further details in the Methods section. The outdoor system operated continuously for 120 h from 6th July to 20th July 2025 (daily from 08:30 to 16:30). Ambient temperatures during operation ranged from 22.3 to 28.1 °C, with outdoor solar irradiance recorded in Supplementary Fig. 30a. Despite fluctuations in photovoltaic output, the energy supply to the reactor remained stable due to the battery buffer (Supplementary Fig. 30b). As shown in Fig. 5c, the

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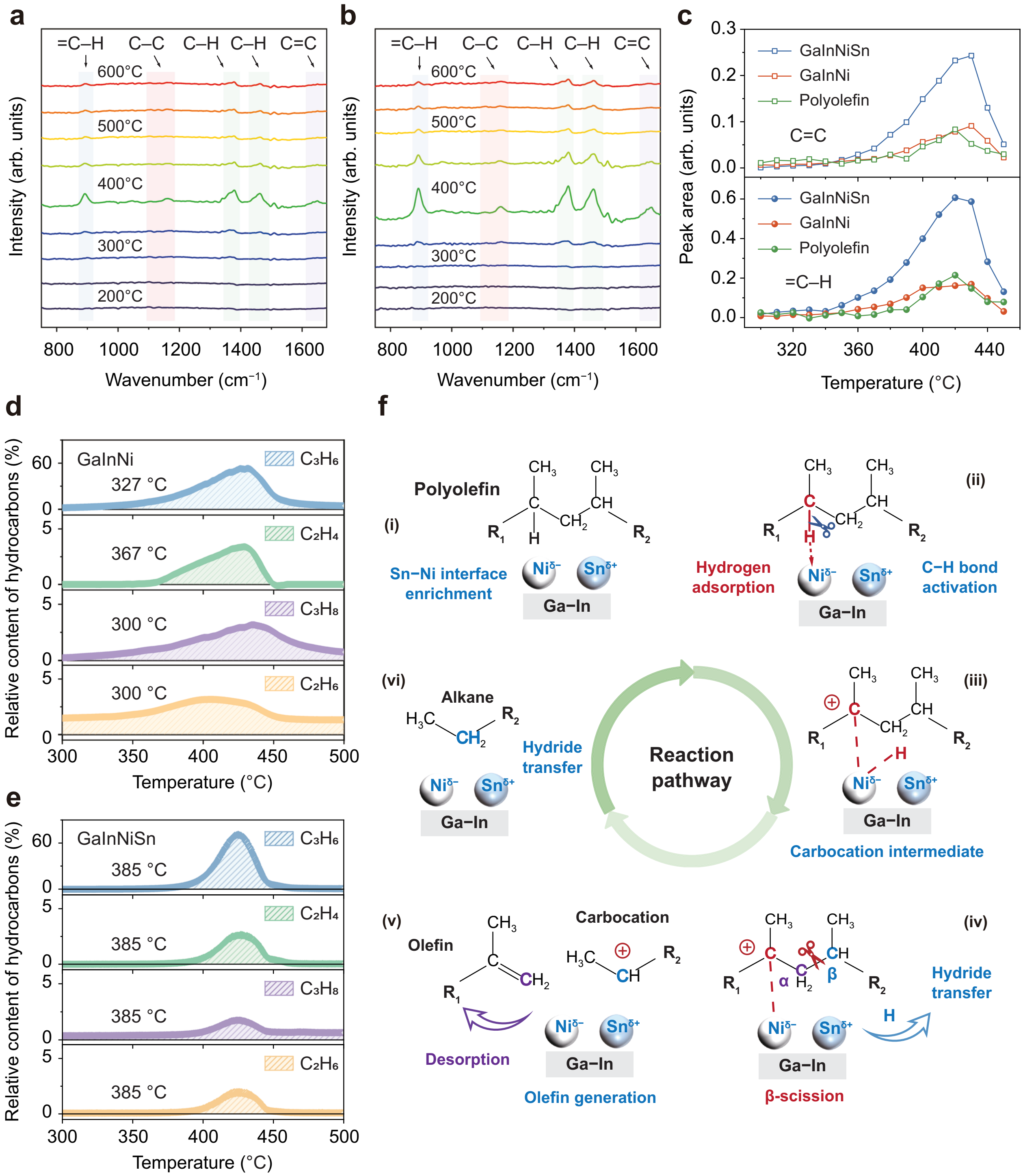

Fig. 4 | Mechanism analysis for polyolefin depolymerization. FTIR spectra of **a** GaInNi LELAC and **b** GaInNiSn MELAC during the catalytic depolymerization of polypropylene. **c** Semi-quantitative spectrum for the peak areas of C = C and =C–H bonds in the catalytic depolymerization process. Online mass spectra of gaseous hydrocarbons during the catalytic depolymerization of polypropylene over **d** GaInNi LELAC and **e** GaInNiSn MELAC (values indicated in the diagram represent the onset temperature). **f** Reaction pathways based on the carbocation mechanism: The fundamental reaction sequence (steps i–vi) involves (i) interface enrichment of $Ni^{\delta-}$–$Sn^{\delta+}$ active sites; (ii) activation of C–H bonds by $Ni^{\delta-}$ sites (which possess strong hydrogen adsorption capacity); (iii) carbocation intermediate formation; (iv) β-scission; (v) generation of olefins and carbocations; and (vi) conversion of carbocations to alkanes via hydrogen transfer.

system achieved a light olefin production rate of 52.8 L h^{-1} and a selectivity of 71.5% (Supplementary Table 10), summing up 6337 L light olefins over 120 hours and thus suggesting excellent operational stability. The energy consumption of the depolymerization process by our system was calculated to be 3.8 kWh $kg_{\text{Light olefins}}^{-1}$. Notably, compared with other depolymerization technologies (Fig. 5d), our system offers over five times greater production rates and ten times longer stability, highlighting its significant advantages[44]. Furthermore, we conducted characterization analysis and performance testing on the recovered GaInNiSn MELAC (Supplementary Figs. 31–33 and Supplementary Tables 11, 12). The results indicate no significant changes occur in surface morphology, elemental composition, and electronic

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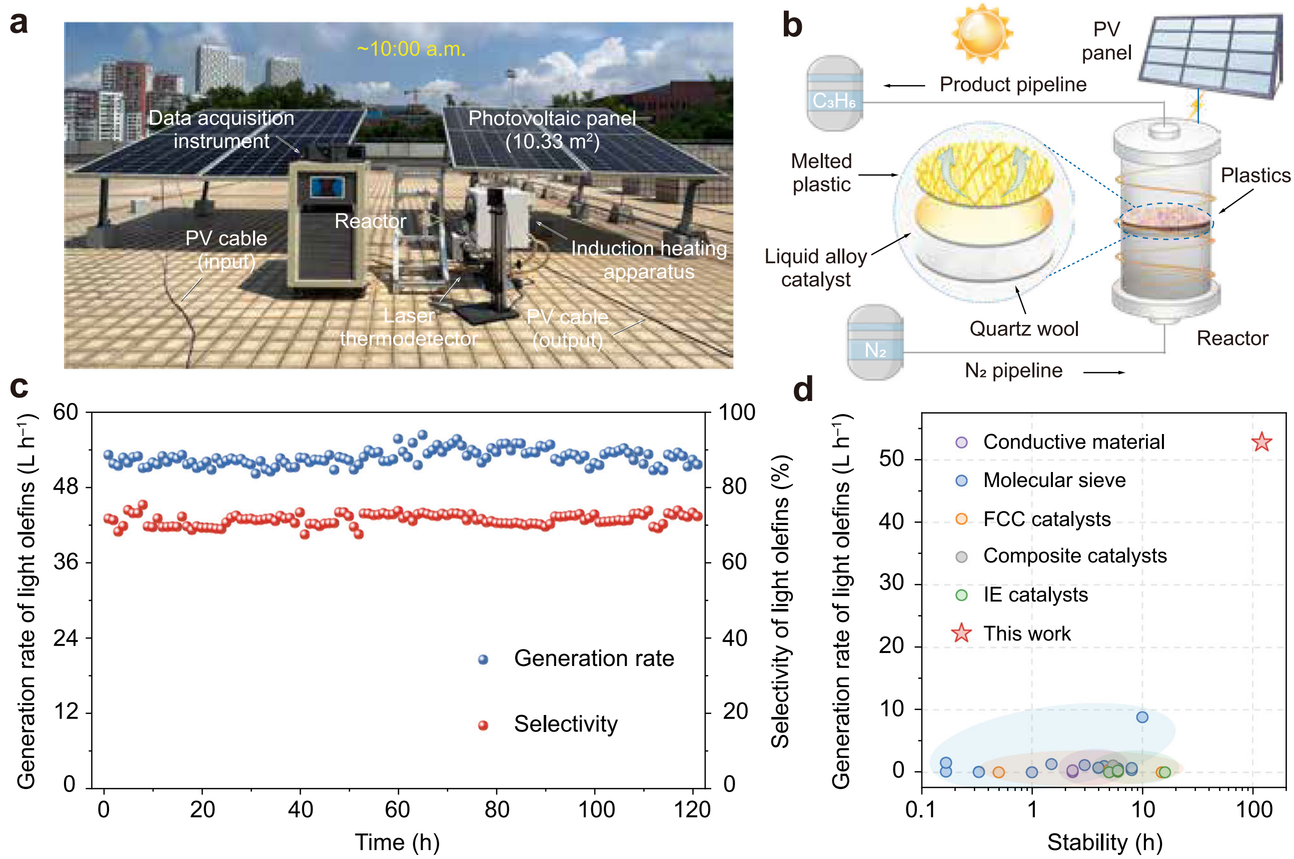

Fig. 5 | Outdoor photovoltaics-driven depolymerization system. a Images of the outdoor system for continuous operation using solar energy and its demonstration configuration. **b** Schematic diagram of the technical route for closed-loop recycling of real-world plastic waste. **c** Generation rate and selectivity of light olefins during 120 h of continuous operation (July 6th to July 20th, 2025; daily from 8:30 to 16:30) using this outdoor system. Reaction conditions: real-world plastic waste fragments (including food packaging bags, take-away containers, drinking water bottle caps, milk bottles, beverage straws, toothpaste tubes, foam boxes, reagent buckets, and medical ziplock bags) are conveyed into the reactor via a continuous feed apparatus at a feed rate of 2.92 g min^{-1}, 11.69 g of GaInNiSn MELAC, N_2 feed gas at a flow rate of 4 mL min^{-1}, and 500 °C for 120 h. **d** Comparison of the performance of different recycling technologies, including operational stability and the generation rate of light olefins[23–26,44,56–69].

structure compared with the fresh GaInNiSn MELAC. Meanwhile, the recovered GaInNiSn MELAC exhibited a light olefin selectivity of 78.1% and a STY of 178.8 mmol g_{cat}^{-1} h^{-1} during the depolymerization of PP plastic. These values are equivalent to the catalytic performance of the fresh GaInNiSn MELAC, demonstrating the recoverability and reusability of the GaInNiSn MELAC.

Discussion

In summary, this work presents a general strategy for depolymerizing real-life polyolefin waste into light olefins, using the GaInNiSn MELAC developed via the interfacial configurational entropy tuning strategy. The depolymerization of pure PP achieved a light olefin STY of 181.5 mmol g_{cat}^{-1} h^{-1} with a selectivity of 79.7%. Also, our strategy exhibited a strong applicability to various pure polyolefins and their blends. Characterization and DFT results unveil that Sn electronically modulates active metal Ni to form $Ni^{\delta-}$–$Sn^{\delta+}$ sites at the GaIn interface, promoting the β-scission depolymerization. Leveraging induction heating for rapid depolymerization, our strategy remarkably inhibits low-temperature side reactions. An outdoor photovoltaics-driven depolymerization system for real-world plastic blends was constructed, maintaining a light olefin production rate of 52.8 L h^{-1} and selectivity of 71.5% during 120 h of continuous operation. Our strategy not only provides a way forward for the efficient and sustainable utilization of plastic waste but also is expected to facilitate the transition to a circular economy.

Methods

Synthesis of liquid alloy catalyst (LAC)

Nitrate salt powders and metallic particles were mixed in a graphite crucible and transferred to a tube furnace. Specifically, for preparing 1.95 g of GaInNiSn MELAC with a targeted element mass ratio of Ga:In:Ni:Sn = 30:15:3:2, the following reagents were accurately weighed as precursors: 1.170 g of metallic gallium granules (Ga), 0.585 g of metallic indium granules (In), 0.078 g of metallic tin granules (Sn), and 0.579 g of nickel nitrate hexahydrate ($Ni(NO_3)_2{\cdot}6H_2O$, corresponding to 0.117 g of Ni). These materials were thoroughly ground in an agate mortar for 5 minutes to ensure homogeneous mixing before being transferred to the graphite crucible. At ambient temperature, a constant flow of 60 mL min^{-1} N_2, regulated by a mass flow controller, was injected into the furnace for 10 min to remove air. The temperature of the LAC (during synthesis) was monitored by a K-type thermocouple controlled by a PID 679 controller. A 30 mL min^{-1} H_2 stream, regulated by another mass flow controller, was then mixed with the 60 mL min^{-1} N_2 stream and introduced into the furnace. The furnace was heated from ambient temperature to 380 °C at a heating rate of 5 °C min^{-1} and maintained at 380 °C for 1.5 h. Subsequently, the LAC was prepared by annealing to 500 °C at a heating rate of 5 °C min^{-1} and subjecting it to slow reduction for 2 h in a mixed atmosphere of H_2 and N_2 (total flow rate of 90 mL min^{-1}, volume ratio of 1:2). After the LAC cooled to room temperature, it was stored in a sealed container.

Catalytic depolymerization

The indoor experimental setup is depicted in Supplementary Fig. 15, comprising a depolymerization reactor, a laser thermo-detector, a thermo-detector control box, an induction heating apparatus (Ambrell EASYHEAT 0224, operating at a frequency of 150 – 400 kHz, with a maximum output of 2.4 kW), and a gas chromatograph. As illustrated in Supplementary Fig. 16, a piece of quartz wool (0.23 g) was placed in a quartz tube with an inner diameter of 16 mm. Specifically, the quartz wool is used to hold the LAC and facilitates uniform gas flow through the reactor; this approach is consistent with standard experimental protocols reported in the literature[31,40]. The LAC used in the depolymerization experiments (including GaIn, GaInSn, GaInNi, and GaInNiSn) all weighed 1.95 g. Unlike supported catalysts, the LAC forms a liquid layer (diameter 1.6 cm, height 0.15 cm) within the reactor (Supplementary Fig. 16c) and thus does not undergo agglomeration during heating. Moreover, the molten LAC cannot permeate into the quartz wool owing to its high surface tension (>500 mN m^{-1}). The micrometer-scale voids formed by the interwoven quartz wool form a capillary structure. For non-wetting interfaces, capillary forces act as repulsive forces, thereby impeding the penetration of the LAC into the quartz wool[45]. Furthermore, the polymer was placed above the LAC and thus did not make direct contact with the quartz wool in depolymerization processing (Supplementary Fig. 16b). As the density of the polymer is lower than that of the LAC, the polymer does not pass through the LAC. Meanwhile, we performed ICP-OES testing on the spent quartz wool after the reactions. The results confirm the absence of elements of the liquid alloy catalyst or polymer within the quartz wool (Supplementary Table 11). The polyolefin plastics used in this manuscript (including LDPE, HDPE, PP, PS, POE, and SBS) exhibited particle sizes in the range of 0.25–0.55 mm (*i.e.*, 30–60 mesh). For real-world plastic waste (*e.g.*, food packaging bags, take-away containers, drinking water bottle caps, milk bottles, beverage straws, toothpaste tubes, foam boxes, reagent buckets, and medical ziplock bags), according to previous studies, we shredded them into fragments with a diameter of less than 2 mm before use, without any further treatment[40]. The lower end of the quartz tube was connected to the nitrogen gas pipeline, and the other end was the gas outlet, which was connected to a gas chromatograph spectrometer (GC, for on-line testing) or to an aluminium foil gas collection bag (for off-line testing). Before the experiment, we calibrated the laser thermometer using a K-type thermocouple to correct systematic errors such as the emissivity of the LAC and quartz tube, the transmission and refraction of the quartz tube, and optical path loss, ensuring accurate laser thermometer readings. The calibrated laser thermometer was employed to measure the temperature of the LAC, with the laser spot position shown in Supplementary Fig. 17a. Furthermore, the temperature rise and fall curve during the induction heating process is shown in Fig S17b. The excellent electrical conductivity of the LAC enabled it to reach the reaction temperature within 5 seconds. After the induction heating apparatus was turned off, the temperature of LAC fell to 250 °C within 30 seconds. For a typical catalytic experiment, nitrogen gas was continuously purged at a flow rate of 20 mL min^{-1} for 10 min to exhaust residual air. A calibrated laser thermometer (measuring wavelength 1.6 µm, temperature range 250 – 1400 °C) was positioned 20 cm away from the reactor to record the LAC temperature, with the output power of the induction heating apparatus adjusted in real time to achieve constant-temperature heating. The cross-sectional area of the LAC was calculated as 37.7 mm^2, significantly larger than the spot area of the laser thermometer (3.14 mm^2). Considering the thermal conductivity of LAC reaches 23.7 W m^{-1}·K^{-1}, it can be thought that the overall temperature of the LAC is uniform under steady state (500 °C). Furthermore, we conducted COMSOL calculations to confirm the highly uniform temperature distribution of the LAC (Supplementary Fig. 18). Consequently, we can measure the temperature of the LAC accurately by positioning the spot of the laser thermometer on the surface of the LAC. Subsequently, a high-frequency electromagnetic induction unit was activated to drive the LAC to depolymerize the plastics. During the 3-minute heating process of the reactor, high-purity nitrogen was continuously introduced as the carrier gas at a flow rate of 4 mL min^{-1}. As illustrated in Supplementary Fig. 15a, the nitrogen gas flowed from the bottom to the top of the quartz tube, ensuring that gaseous products formed were immediately swept away by the nitrogen stream and transferred to the gas collection bag, rather than remaining in continuous contact with the LAC. After the reaction proceeded for 3.0 min, the electromagnetic induction device was deactivated. Once the reactor was cooled to ambient temperature, the LAC was recovered with a pipette and sealed in a reagent bottle containing 2 wt% acetone for subsequent testing (Supplementary Figs. 31, 32 and Supplementary Tables 11, 12). The recovered LAC can be reused after undergoing ultrasonic cleaning at 120 watts for 20 minutes (Supplementary Fig. 33). The conversion of plastic, the selectivity of light olefins, and the STY of light olefins were calculated using the following equations:

$$\text{Conversion} = \left(\frac{M_{\text{Reactor},0} - M_{\text{Reactor},t}}{M_{\text{Plastic}}}\right) \times 100\% \tag{1}$$

$$\text{Selectivity} = \left(\frac{X_i}{\sum_1^n X_i}\right) \times 100\% \tag{2}$$

$$\text{STY} = \frac{\sum n_{\text{Lightolefins}}}{M_{\text{Liquid alloy}} \times t} \tag{3}$$

where $M_{\text{Reactor,0}}$ denotes the mass of the reactor before reaction (comprising the quartz tube, quartz wool, liquid alloy catalyst, and plastic), $M_{\text{Reactor,t}}$ denotes the mass of the reactor after reaction, and M_{Plastic} denotes the mass of plastic added to the reactor before reaction; X_i denotes the molar fraction of gas product i (including H_2, CH_4, C_2H_4, C_2H_6, C_3H_6, C_3H_8, C_4H_8, and C_4H_{10}), and n is the number of gas products; $\sum n_{\text{Light olefins}}$ is the amount of light olefins (including C_2H_4, C_3H_6, and C_4H_8) in millimoles, $M_{\text{Liquid alloy}}$ is the mass of the LAC in grams, and t is the reaction time in hours. The detailed conversion and mass balance analysis for the depolymerization reaction of polypropylene plastic is recorded in Supplementary Tables 13, 14.

Application of an outdoor photovoltaics-driven depolymerization system

The outdoor photovoltaics-driven depolymerization system built in this study consisted of two components: a solar power system and a waste plastic depolymerization system. The solar power system used four TY-450M photovoltaic panels (2278 × 1134 × 30 mm) with an irradiation area of 10.3 m^2 (module efficiency of 24%). Additionally, four 12 V – 200 Ah storage batteries (storage capacity of 9.6 kWh) were equipped to store photovoltaic power, which is then converted by inverters for use in depolymerization reactions. Meanwhile, the batteries also stabilize power output and balance the effects of variations in weather and sunlight intensity[46]. The waste plastic depolymerization system used a customized vacuum heat pipe reactor, which comprised an outer pipe of 300 mm in length and 32 mm in diameter, an inner pipe of 450 mm in length and 20 mm in diameter, and a vacuum layer of 300 mm in length and 3 mm in thickness (Supplementary Fig. 29a). The vacuum heat pipe reactor was fixed using an aluminium bracket (dimensions: 330 × 350 × 650 mm). Subsequently, post-consumer plastic product fragments (e.g., food packaging bags, take-away containers, drinking water bottle caps, milk bottles, beverage straws, toothpaste tubes, foam boxes, reagent buckets, and medical ziplock bags) were loaded into the continuous feed apparatus and conveyed to the reactor via a screw rod at the left end of the apparatus (Supplementary Fig. 29b). The continuous feed apparatus delivers plastic fragments into the inner tube at a feed rate of 2.92 g min^{-1}. For the production of light olefins (C_2H_4, C_3H_6, and C_4H_8), 11.7 g of the GaInNiSn MELAC was used. The GaInNiSn MELAC was placed into the reactor only before the commencement of the reaction and was neither added nor removed during continuous operation. The induction heating apparatus was a JRHG-3 KW UHF ultra-high frequency induction heating machine (maximum output power: 3 kW; working frequency: 500 − 1100 kHz) supplied by Jinrui Intelligent Technology Co., Ltd. (Dongguan, Guangdong Province, China). The outdoor system was constructed on the top floor of the experimental building and operated continuously for over 120 hours between July 6th and July 20th, 2025 (daily from 8:30 to 16:30), with a reaction temperature of 500 °C and an ambient temperature ranging from 22.3 to 28.1 °C (Supplementary Fig. 30). The generated light olefins were collected in aluminium foil gas sampling bags, and their detailed composition and corresponding percentages were determined via GC-MS analysis (offline mode).

First-principles calculations

Considering that GaInNi and GaInNiSn remained in a liquid state throughout the depolymerization process, ab initio molecular dynamics (AIMD) simulations were performed to obtain reasonable equilibrium configurations for both systems. Before the AIMD simulation, 100 atoms with target compositions were randomly generated for each alloy using PACKMOL in a 13 Å × 13 Å × 25 Å periodic box. To eliminate spurious periodic interactions between adjacent systems and accommodate the thermal mobility of liquid atoms, a 15 Å vacuum layer was added along the height direction of the 25 Å box, resulting in a slab model of the final periodic box size of 13 Å × 13 Å × 40 Å for the formal AIMD simulations. The target compositions of the two alloy systems were as follows: GaInNi (Ga:In:Ni = 67:19:14) and GaInNiSn (Ga:In:Ni:Sn = 69:20:8:3). Notably, all these details (the number of atoms and the cell dimensions) are now explicitly stated in the corresponding figure captions for the AIMD snapshots. For the AIMD simulations, the plane-wave cutoff energy was set to 520 eV; a 1 × 1 × 1 k-mesh was used to sample the Brillouin zone, and spin-orbit coupling (SOC) was taken into account in the calculations[47]. The samples were then equilibrated at the equilibrium temperature (773 K) for 10 ps in the canonical ensemble (NVT). The time step was set to 2 fs, and the integration of Newton's equation used the Verlet algorithm implemented in the Vienna Ab-initio Simulation Package (VASP). Subsequently, suitable atomic configurations with active sites proximal to the surface were selected from the 10 ps AIMD trajectories, and periodic plane-wave density functional theory (DFT) calculations were performed on GaInNi and GaInNiSn using the VASP (VASP 6.3.2 optcell)[48]. To investigate the activation mechanism of C−H bonds, the H adsorption energy on the active Ni sites in the two systems was calculated. The projector augmented plane wave (PAW) pseudopotential basis set and the generalized gradient approximation (GGA) functional with the Perdew−Burke−Ernzerhof (PBE) parametrization were employed in these calculations. A kinetic energy cutoff of 520 eV for plane-wave expansion was used for the two systems. Brillouin zone sampling was performed using a 3 × 1 × 3 k-point grid centered at the Γ point. The force convergence criterion was set to 0.02 eV $Å^{-1}$, and spin-orbit coupling effects were considered in the calculations. A 15 Å vacuum layer was introduced to avoid the influence of adjacent crystal planes on surface atoms, and the top atomic layers (surface region) were fully relaxed.

Data availability

The data supporting the findings of the study are included in the main text, supplementary information, and source data files. Source data are provided with this paper. All data are available from the corresponding author upon request. Source data are provided with this paper.

References

1. Häussler, M., Eck, M., Rothauer, D. & Mecking, S. Closed-loop recycling of polyethylene-like materials. *Nature* **590**, 423–427 (2021).
2. Zhao, J. et al. Highly selective upcycling of plastic mixture waste by microwave-assisted catalysis over Zn/b-ZnO. *Nat. Commun.* **16**, 1726 (2025).
3. Geyer, R., Jambeck, J. R. & Law, K. L. Production, use, and fate of all plastics ever made. *Sci. Adv.* **3**, e1700782 (2017).
4. Kang, H. X. et al. Mining the carbon intermediates in plastic waste upcycling for constructing C–S bond. *J. Am. Chem. Soc.* **146**, 18639–18649 (2024).
5. Zheng, K. et al. Progress and perspective for conversion of plastic wastes into valuable chemicals. *Chem. Soc. Rev.* **52**, 8–29 (2023).
6. Xiang, M. et al. Triple kill: Fabrication of composites coming from waste face masks, polystyrene microplastics, graphene, and their electromagnetic interference shielding behaviors. *Carbon Neutraliza* **2**, 616–628 (2023).
7. Han, Z. Y. et al. Recycling waste rubber into single-walled carbon nanotubes: Narrow chirality distribution and hydrogen byproduct. *Carbon Neutraliza* **4**, e70059 (2025).
8. Zhang, S. B. et al. Depolymerization of polyesters by a binuclear catalyst for plastic recycling. *Nat. Sustain* **6**, 965–973 (2023).
9. Williams, J. M. et al. Hybrid thermo-electrochemical conversion of plastic wastes commingled with marine biomass to value-added products using renewable energy. *Energ. Environ. Sci.* **16**, 5805–5821 (2023).
10. Du, J. J. et al. Efficient solvent- and hydrogen-free upcycling of high-density polyethylene into separable cyclic hydrocarbons. *Nat. Nanotechnol.* **18**, 772–779 (2023).
11. Hu, P. et al. Stable interfacial ruthenium species for highly efficient polyolefin upcycling. *J. Am. Chem. Soc.* **146**, 7076–7087 (2024).
12. Cao, R. C., Xiao, D. Q., Wang, M., Gao, Y. J. & Ma, D. Solar-driven photocatalysis for recycling and upcycling plastics. *Appl Catal. B-Environ.* **341**, 123357 (2024).
13. Wimberger, L., Ng, G. & Boyer, C. Light-driven polymer recycling to monomers and small molecules. *Nat. Commun.* **15**, 2510 (2024).
14. Wang, X. Y., Gao, Y. S. & Tang, Y. Sustainable developments in polyolefin chemistry: Progress, challenges, and outlook. *Prog. Polym. Sci.* **143**, 101713 (2023).
15. Sun, J. K. et al. Catalytic upcycling of polyolefins. *Chem. Rev.* **124**, 9457–9579 (2024).
16. Zhang, W. et al. Low-temperature upcycling of polyolefins into liquid alkanes via tandem cracking-alkylation. *Science* **379**, 807–811 (2023).
17. Wang, H. S. et al. Visible light-triggered depolymerization of commercial polymethacrylates. *Science* **387**, 874–880 (2025).
18. Gao, R. L., Mao, S. J., Lu, B., Liu, W. C. & Wang, Y. Efficient upcycling of polyolefin waste to light sromatics via coupling C-C scission and carbonylation. *Angew. Chem. Int Ed.* **64**, e202424334 (2025).
19. Kang, Q. Y. et al. Entropy confinement promotes hydrogenolysis activity for polyethylene upcycling. *Angew. Chem. Int Ed.* **62**, e202313174 (2023).
20. Cen, Z. Y. et al. Upcycling of polyethylene to gasoline through a self-supplied hydrogen strategy in a layered self-pillared zeolite. *Nat. Chem.* **16**, 871–880 (2024).
21. Miao, Y. X. et al. Photothermal recycling of waste polyolefin plastics into liquid fuels with high selectivity under solvent-free conditions. *Nat. Commun.* **14**, 4242 (2023).
22. Feng, J. Y. et al. Micelles cascade assembly to tandem porous catalyst for waste plastics upcycling. *Angew. Chem. Int Ed.* **63**, e202405252 (2024).
23. Dong, Q. et al. Depolymerization of plastics by means of electrified spatiotemporal heating. *Nature* **616**, 488–494 (2023).

24. Selvam, E., Yu, K. W., Ngu, J., Najmi, S. & Vlachos, D. G. Recycling polyolefin plastic waste at short contact times via rapid joule heating. *Nat. Commun.* **15**, 5662 (2024).
25. Conk, R. J. et al. Catalytic deconstruction of waste polyethylene with ethylene to form propylene. *Science* **377**, 1561–1566 (2022).
26. Conk, R. J. et al. Polyolefin waste to light olefins with ethylene and base-metal heterogeneous catalysts. *Science* **385**, 1322–1327 (2024).
27. Ma, Y. J. et al. High-entropy approach traditional doping strategy for layered oxide cathodes in alkali-metal-ion batteries: A comparative study. *Energy Storage Mater.* **79**, 104295 (2025).
28. Yao, Y. G. et al. High-entropy nanoparticles: Synthesis-structure-property relationships and data-driven discovery. *Science* **376**, eabn3103 (2022).
29. Huang, M. S. et al. Configuration-entropy-driven electrolyte with anion-enhanced solvation structures for fast-charging sodium metal batteries. *Adv Funct Mater*, e24686, https://doi.org/10.1002/adfm.202524686 (2025).
30. Upham, D. C. et al. Catalytic molten metals for the direct conversion of methane to hydrogen and separable carbon. *Science* **358**, 917–920 (2017).
31. Chen, L. N. et al. Ternary NiMo-Bi liquid alloy catalyst for efficient hydrogen production from methane pyrolysis. *Science* **381**, 857–861 (2023).
32. Kelly, S. Introducing the basics of modeling EXAFS spectra in Artemis, http://xrayabsorption.org/tutorials/ (2009).
33. Qin, J. B. et al. Advanced catalysts for the chemical recycling of plastic waste. *Adv. Mater.* **37**, 202418138 (2025).
34. Dong, Q., Hu, S. & Hu, L. Electrothermal synthesis of commodity chemicals. *Nat. Chem. Eng.* **1**, 680–690 (2024).
35. Yang, H. M. et al. Distributed electrified heating for efficient hydrogen production. *Nat. Commun.* **15**, 3868 (2024).
36. Moritz, M. et al. Supported catalytically active liquid metal solutions: Liquid metal catalysis with ternary alloys, enhancing activity in propane dehydrogenation. *Acs Catal.* **14**, 6440–6450 (2024).
37. Clarke, R. W. et al. Dynamic crosslinking compatibilizes immiscible mixed plastics. *Nature* **616**, 731–739 (2023).
38. Shi, C. X., Diment, W. T. & Chen, E. Y. X. Closed-loop recycling of mixed plastics of polyester and CO_2-based polycarbonate to a dingle monomer. *Angew. Chem. Int Ed.* **63**, e202405083 (2024).
39. Xu, Z. et al. Chemical upcycling of polyethylene, polypropylene, and mixtures to high-value surfactants. *Science* **381**, 666–671 (2023).
40. Wang, M. et al. Complete hydrogenolysis of mixed plastic wastes. *Nat. Chem. Eng.* **1**, 376–384 (2024).
41. Hu, P. et al. Identifying the active phase on atomically dispersed catalysts for propane dehydrogenation: Positively charged vs metallic transition metals. *Acs Catal.* **14**, 8602–8618 (2024).
42. Bercaw, J. E., Hazari, N., Labinger, J. A., Scott, V. J. & Sunley, G. J. Selective methylative homologation: An alternate route to alkane upgrading. *J. Am. Chem. Soc.* **130**, 11988–11995 (2008).
43. Yu, S. M. et al. Alkane activation initiated by hydride transfer: Co-conversion of propane and methanol over H-ZSM-5 zeolite. *Angew. Chem. Int Ed.* **54**, 7363–7366 (2015).
44. Elordi, G., Olazar, M., Artetxe, M., Castaño, P. & Bilbao, J. Effect of the acidity of the HZSM-5 zeolite catalyst on the cracking of high density polyethylene in a conical spouted bed reactor. *Appl Catal. a-Gen.* **415**, 89–95 (2012).
45. Yun, F. F. et al. Voltage-induced penetration effect in liquid metals at room temperature. *Natl. Sci. Rev.* **7**, 366–372 (2020).
46. Wang, P. X. et al. Rapid-Response Electromagnetic Metamaterial Enabling Round-The-Clock Electrified CO_2 Methanation. *Adv Funct Mater*, e24133, https://doi.org/10.1002/adfm.202524133 (2025).
47. Monkhorst, H. J. & Pack, J. D. Special points for Brillouin-Zone integrations. *Phys. Rev. B* **13**, 5188–5192 (1976).
48. Kresse, G. & Furthmuller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169–11186 (1996).
49. Krishnamurthi, V. et al. Structural evolution of liquid metals and alloys. *Adv. Mater.* **36**, 2403885 (2024).
50. Polo-Garzon, F. et al. Low-temperature dechlorination of polyvinyl chloride (PVC) for production of H_2 and carbon materials using liquid metal catalysts. *Sci. Adv.* **10**, eadm9963 (2024).
51. Zuraiqi, K. et al. Unveiling metal mobility in a liquid Cu-Ga catalyst for ammonia synthesis. *Nat. Catal.* **7**, 1044–1052 (2024).
52. Gao, J. Y. et al. Microwave-powered liquid metal degradation of polyolefins. *Adv. Mater.* **37**, 2412539 (2025).
53. Zhai, Z. Z. et al. Electromagnetically heating and oscillating liquid metal for catalyzing polyester depolymerization. *Adv. Mater.* **37**, 2502011 (2025).
54. Tang, J. M. et al. Molten Sn solvent expands liquid metal catalysis. *Nat. Commun.* **16**, 907 (2025).
55. Zhang, H. R. et al. Improving the methane oxidation by self-adaptive optimization of liquid-metal catalysts. *Angew. Chem. Int Ed.* **64**, e202421554 (2025).
56. Lin, Y. H., Sharratt, P. N., Garforth, A. A. & Dwyer, J. Catalytic conversion of polyolefins to chemicals and fuels over various cracking catalysts. *Energy Fuels* **12**, 767–774 (1998).
57. Lovás, P., Hudec, P., Jambor, B., Hájeková, E. & Hornácek, M. Catalytic cracking of heavy fractions from the pyrolysis of waste HDPE and PP. *Fuel* **203**, 244–252 (2017).
58. Lin, Y. H. & Yang, M. H. Catalytic conversion of commingled polymer waste into chemicals and fuels over spent FCC commercial catalyst in a fluidised-bed reactor. *Appl Catal. B-Environ.* **69**, 145–153 (2007).
59. Lin, Y. H. & Yang, M. H. Catalytic pyrolysis of polyolefin waste into valuable hydrocarbons over reused catalyst from refinery FCC units. *Appl Catal. a-Gen.* **328**, 132–139 (2007).
60. Olazar, M. et al. Influence of FCC catalyst steaming on HDPE pyrolysis product distribution. *J. Anal. Appl Pyrol* **85**, 359–365 (2009).
61. Eschenbacher, A. et al. Boron-modified mesoporous ZSM-5 for the conversion of pyrolysis vapors from LDPE and mixed polyolefins: Maximizing the C_2–C_4 olefin yield with minimal carbon footprint. *Acs Sustain Chem. Eng.* **9**, 14618–14630 (2021).
62. Eschenbacher, A., Varghese, R. J., Abbas-Abadi, M. S. & Van Geem, K. M. Maximizing light olefins and aromatics as high value base chemicals via single step catalytic conversion of plastic waste. *Chem. Eng. J.* **428**, 132087 (2022).
63. Dong, Z. W. et al. Efficient conversion of polyethylene to light olefins by self-confined cracking and reforming. *Nat. Commun.* **16**, 7964 (2025).
64. Tianrui, Bi. et al. Closed-loop recycling of polyethylene to ethylene and propylene via a kinetic decoupling–recoupling strategy. *Nat. Chem. Eng.* **2**, 650–661 (2025).
65. Zhao, Y. Z. et al. Catalytic cracking of polypropylene by using Fe-SBA-15 synthesized in an acid-free medium for production of light hydrocarbon oils. *J. Anal. Appl Pyrol* **146**, 104755 (2020).
66. Scott, D. S., Majerski, P., Piskorz, J., Radlein, D. & Barnickel, M. Production of liquid fuels from waste plastics. *Can. J. Chem. Eng.* **77**, 1021–1027 (1999).
67. Encinar, J. M. & González, J. F. Pyrolysis of synthetic polymers and plastic wastes. Kinetic study. *Fuel Process. Technol.* **89**, 678–686 (2008).
68. Duan, J. D. et al. Coking-resistant polyethylene upcycling modulated by zeolite micropore diffusion. *J. Am. Chem. Soc.* **144**, 14269–14277 (2022).

69. Wang, N. M. et al. Chemical recycling of polyethylene by tandem catalytic conversion to propylene. *J. Am. Chem. Soc.* **144**, 18526–18531 (2022).

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Author contributions
C.X., X.Z., D.T., and B.J. conceived and designed the research; C.X., J.L., H.W., X.W., and Q.Z. conducted the experiments and analyzed the data; H.Y. performed the DTF calculations; C.X., P.W., and J.M. constructed the outdoor system; C.X. and B.J. visualized the data and drafted the manuscript; X.Z., D.T., and B.J. supervised the research; C.X. and B.J. revised and edited the manuscript. All authors discussed the results and commented on the manuscript.

Competing interests
The authors declare no competing interests.

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